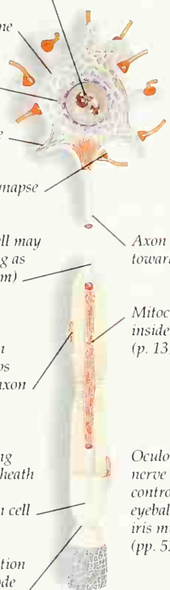
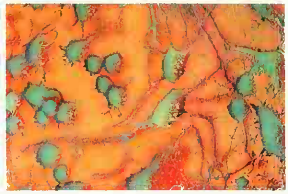


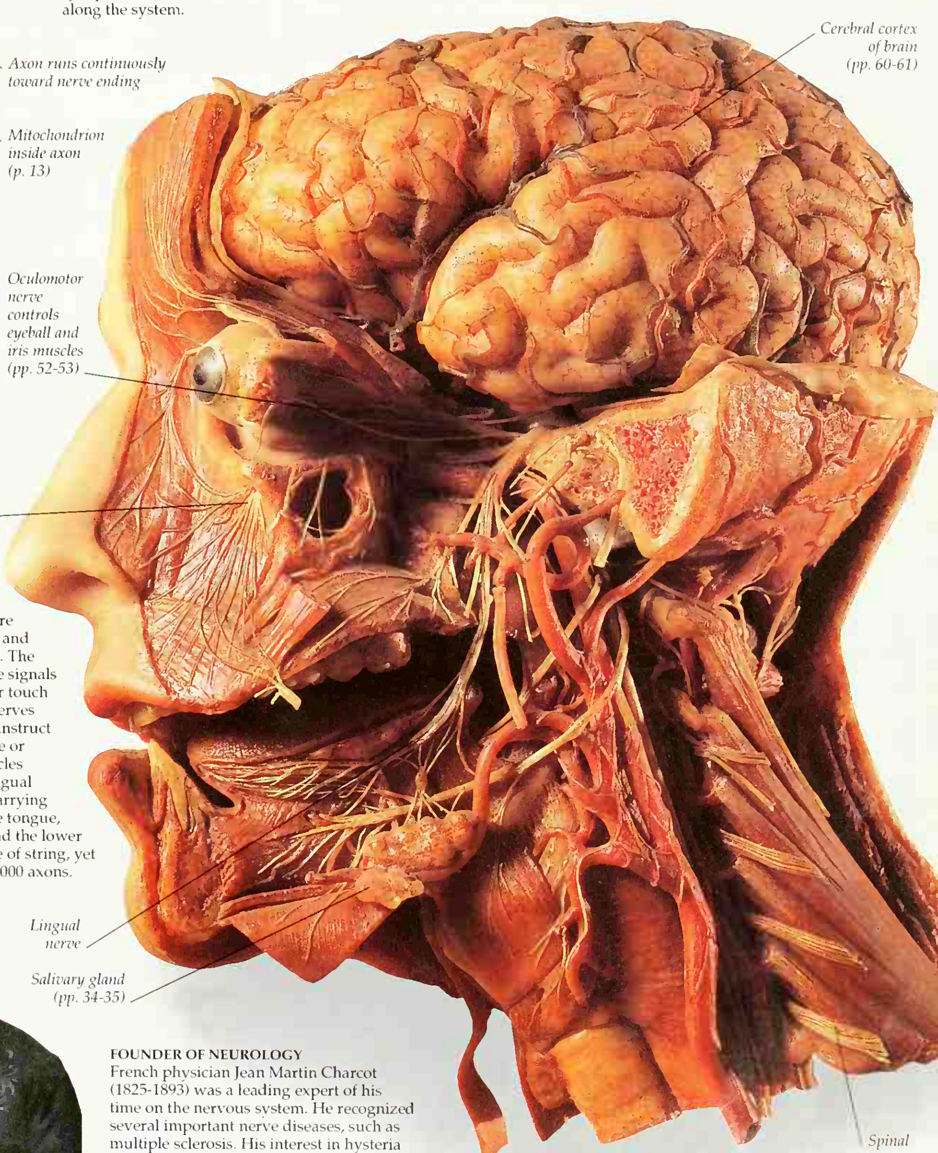
THE INDIVIDUAL NERVE CELL

The basic unit of the nervous system is the nerve cell, or neuron. Its main body is much the same as any other cell. It also has many small branched endings called dendrites, and one wirelike part that is much longer, called an axon. Electrical nerve signals are received from other neurons across tiny gaps – synapses – into the dendrites. The signals travel along the cell's membrane and into the axon. In turn, this forms synapses with more neurons farther along the system.



BRANCHES EVERYWHERE

The electron microscope shows neurons (green) distinguished from their surrounding cells, showing how their branches form an immense interconnected net. One neuron can have links with more than 1,000 others. Nerve signals can take any path between them, and the possible routes are countless.



FACIAL NERVES

This model shows just a few of the nerves in the face and neck. Most are a combination of sensory and motor, in bundles of cells. The sensory nerves take nerve signals from sight, smell, taste, or touch to the brain. The motor nerves transmit brain signals to instruct the facial muscles to smile or frown, and the neck muscles to move the head. The lingual nerve is solely sensory, carrying touch sensations from the tongue, the floor of the mouth, and the lower jaw. It is as thin as a piece of string, yet it contains as many as 10,000 axons.

FOUNDER OF NEUROLOGY

French physician Jean Martin Charcot (1825-1893) was a leading expert of his time on the nervous system. He recognized several important nerve diseases, such as multiple sclerosis. His interest in hysteria and hypnosis opened the way for the branch of medicine known as psychiatry. Among his students was Sigmund Freud (p. 63).